Analysing Customer Reviews using NLP: From Text to Insight

Problem Statement:

You are tasked with building a complete NLP analysis pipeline on a dataset of Amazon Beauty product reviews. Your goal is to extract meaningful insights from raw text by identifying the topics customers discuss, the brands and products they mention, and the sentiment they express. The pipeline must follow a logical analytical progression: clean the text, represent it in multiple ways, discover structure within it, and finally model and explain it.

Dataset:

You have been provided with a CSV file: beauty_reviews.csv. It contains real Amazon Beauty product reviews with exactly two columns:

review_text, sentiment

Where review_text is the raw review string and sentiment is a binary label (1 = positive, 0 = negative).

Evaluate your classifier on a held-out test split of at least 20% using random_state=42.

The pipeline should:

- **Preprocess the text** using two separate token streams: (a) a POS-filtered stream (NOUN and ADJ only, lemmatised) for TF-IDF vectorisation and LDA topic modelling, and (b) a lightly cleaned stream (all content words, stopwords removed but no POS filter) for Word2Vec training.
- **Apply Named Entity Recognition (NER)** on the full unfiltered text to extract brands, products, and locations. Report the top 20 most-mentioned entities using a bar chart. Compute, for each entity in the top 20, the mean sentiment score of reviews that mention it to understand which entities are associated with positive or negative feedback.
- **Vectorise using TF-IDF** (scikit-learn) on the POS-filtered tokens. Validate the vectorisation by computing a pairwise cosine similarity heatmap on a random sample of reviews (e.g., ~50) and check that reviews with the same sentiment label show higher similarity than cross-label pairs; annotate the heatmap axes with sentiment labels to make this visible.
- **Train Word2Vec** model (gensim) on the lightly-cleaned tokens and average word vectors to create document embeddings. Generate 2-D PCA scatter plots for both TF-IDF (after TruncatedSVD to 50 dims) and Word2Vec embeddings, coloured by sentiment, and compare the cluster structures to understand what each representation captures.

- **Discover topics using LDA** on a Bag-of-Words corpus (gensim Dictionary + doc2bow) built from POS-filtered tokens (not TF-IDF). Train models for k = 5, 7, and 10, select the best using coherence score (c_v), generate word cloud panels for each topic, and assign human-readable labels. For each review, identify its dominant topic (highest probability assignment), then compute and report in a table the proportion of positive vs negative reviews for each topic, linking the discovered topics to sentiment labels.
- **Train three sentiment classifiers** on TF-IDF features: MultinomialNB, BernoulliNB, and Logistic Regression. Evaluate all three using accuracy, precision, recall, F1-score, and plot their ROC curves on the same axes. Present results in a single comparison table.
- **Optionally:** Use a pre-trained DistilBERT model (HuggingFace Transformers) to generate contextual document embeddings by tokenising each review and extracting the [CLS] token from the last hidden state. Train a Logistic Regression classifier on these embeddings and include it as a fourth row in the classifier comparison table. In the report, reflect on whether any performance gains justify the additional computational cost and discuss the role of domain-specific pre-training for beauty product reviews.

Constraints:

- Use spaCy (`en_core_web_lg`) for tokenisation, POS tagging, lemmatisation, and NER.
- TF-IDF via scikit-learn; Word2Vec via gensim; LDA via gensim's `LdaModel` with a BoW corpus - not a TF-IDF matrix.
- Maintain two token streams: POS-filtered (NOUN + ADJ) for TF-IDF and LDA; lightly-cleaned (all content words) for Word2Vec.
- LDA coherence scored using gensim's `CoherenceModel` with `coherence='c_v'`; k explored over {5, 7, 10}.
- All visualisations via matplotlib or seaborn. Each stage of the pipeline must be a separate, well-commented function.
- Wrap all execution logic inside `if __name__ == '__main__'`.

Deliverable / Submission Format:

- **Solution 1:** A clean, well-commented Python script (.py).
- **Solution 2:** A short written report (PDF, max 2 pages) covering: (a) rationale for your two-stream preprocessing design, (b) top 5 LDA topics with human-readable labels and their positive/negative sentiment split, (c) a three-row classifier comparison table (MultinomialNB, BernoulliNB, Logistic Regression) showing accuracy, F1, and AUC, and (d) one insight linking NER entities to sentiment (e.g. which brand appears most in negative reviews).

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